

α -collision threshold sweep

Date: 2026-05-21 · **Scope:** 4 model configurations × 2 benchmarks × 4 α arms

Quick reference for the α -sweep results across the instruct models we've completed end-to-end, plus Qwen3-8B with thinking disabled as the decisive-test control for the thinking-vs-saturation question. All numbers below are extracted live from the metrics JSONs at the paths cited beneath each table; every plot is rendered from the same data.

Scope

Dimension	Value
Model configurations	Qwen2.5-Coder-7B-Instruct, CodeLlama-7B-Instruct, OpenCodeInterpreter-DS-1.3B, Qwen3-8B-NoThink
Datasets	MBPP, HumanEval (MBPP-500, HumanEval-164)
α grid	2.0, 2.5, 3.0, 5.0
Temperature	T=1.0 (fixed; α is the sweep parameter)
Samples per task	10

Metrics shown: pass@k for k in {1, 3, 5, 10} (Chen et al. 2021 unbiased estimator) and CodeBLEU pairwise diversity.

CodeLlama-7B-Instruct — HumanEval

n_tasks: 164 · **samples/task:** 10

α	pass@1	pass@3	pass@5	pass@10	codebleu_div	NAUADC
2.0	27.74%	30.79%	31.73%	32.32%	0.0566	1.0175
2.5	25.85%	35.04%	38.08%	40.85%	0.1606	1.0144
3.0	25.24%	35.82%	40.03%	44.51%	0.2381	1.0134
5.0	24.82%	36.75%	41.45%	46.95%	0.2804	1.0236

Source dir: results/pless_alpha_full_humaneval/codellama--CodeLlama-7b-Instruct-hf/humaneval/metrics/

Files: pless_alpha_a{2.0,2.5,3.0,5.0}_t1.0_metrics.json

CodeLlama-7B-Instruct — MBPP

n_tasks: 500 · **samples/task:** 10

α	pass@1	pass@3	pass@5	pass@10	codebleu_div	NAUADC
2.0	41.78%	43.25%	43.71%	44.20%	0.0677	1.0085
2.5	41.24%	45.87%	47.50%	49.20%	0.1920	1.0446
3.0	40.66%	46.64%	48.82%	50.80%	0.2354	1.0770
5.0	40.32%	48.10%	50.70%	53.20%	0.3042	1.1186

Source dir: results/pless_alpha_full_mbpp/codellama--CodeLlama-7b-Instruct-hf/metrics/

Files: pless_alpha_a{2.0,2.5,3.0,5.0}_t1.0_metrics.json

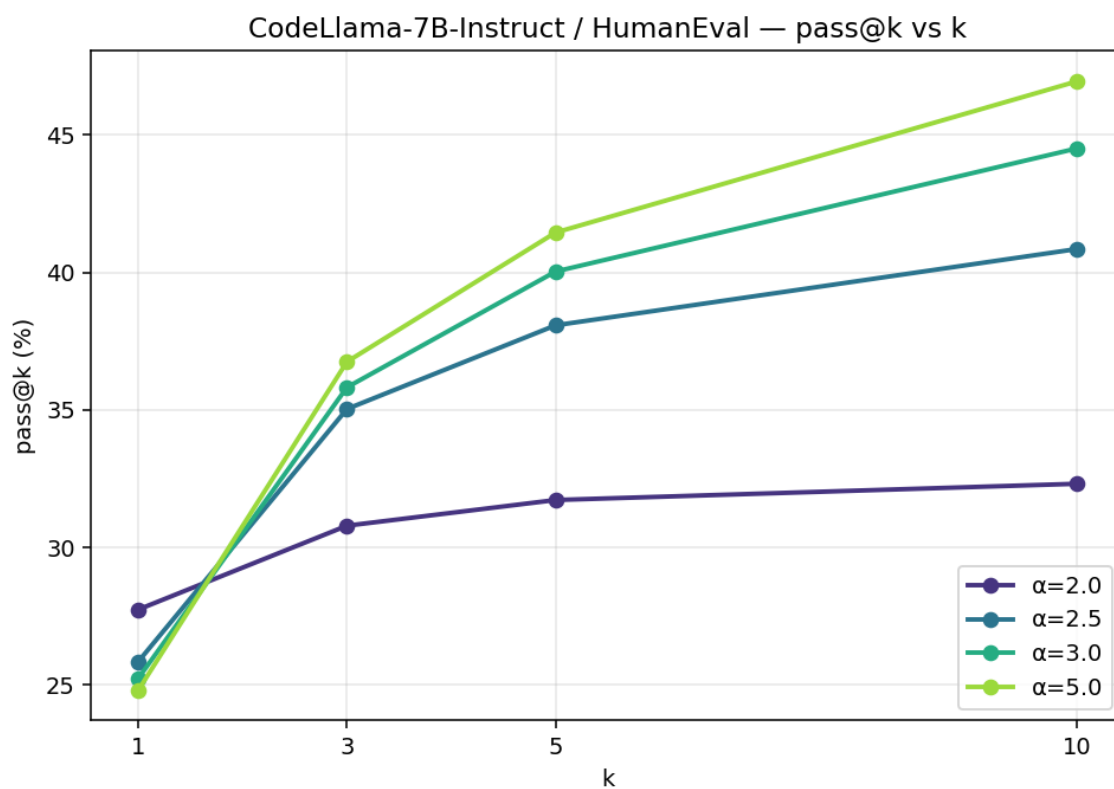


Figure 1: pass@k vs k

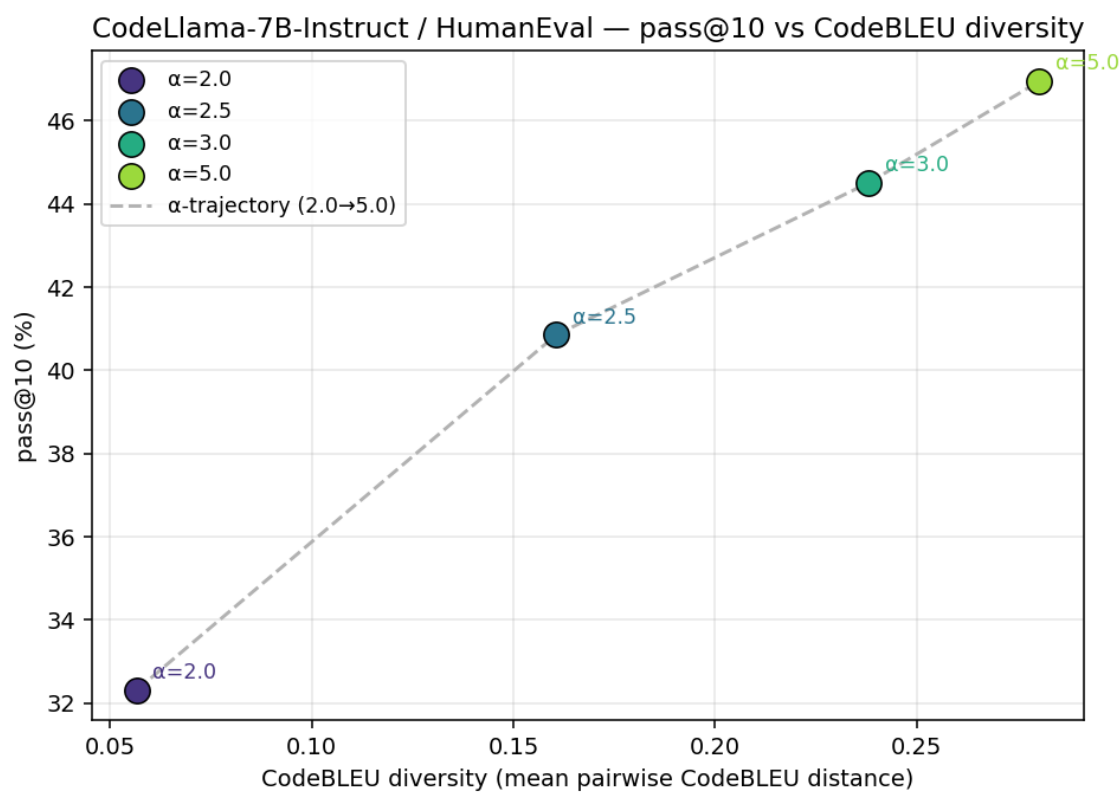


Figure 2: pass@10 vs CodeBLEU diversity

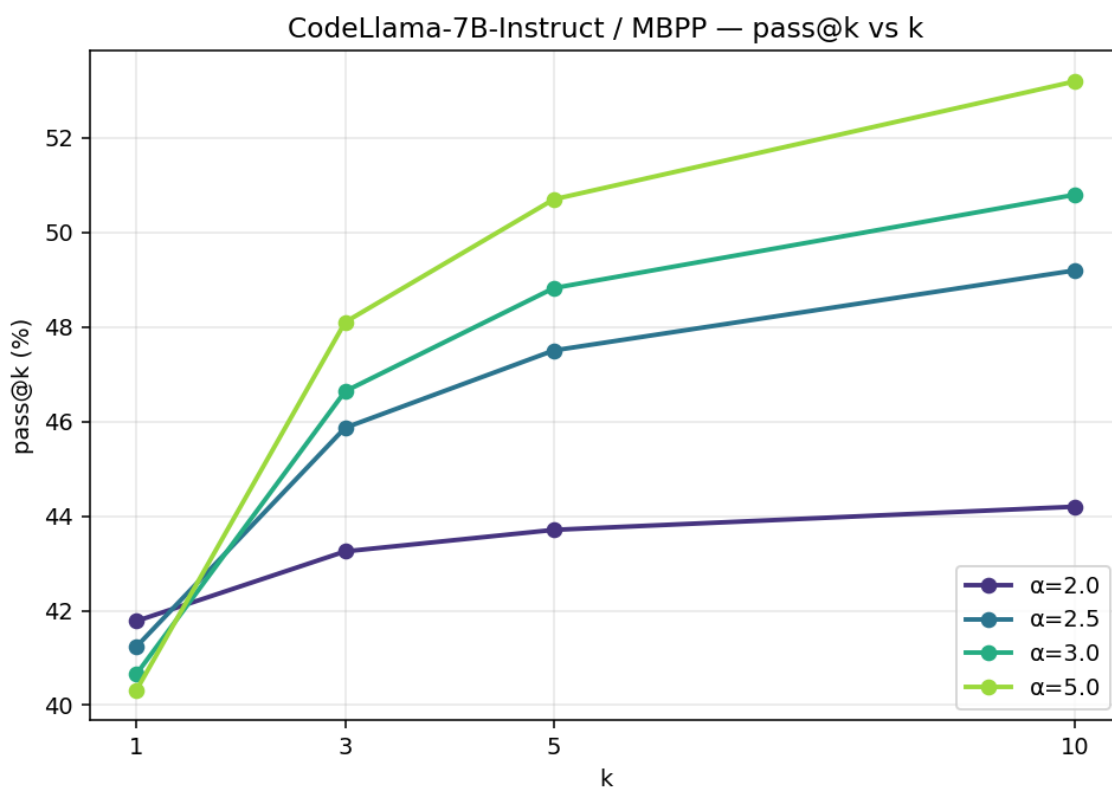


Figure 3: pass@k vs k

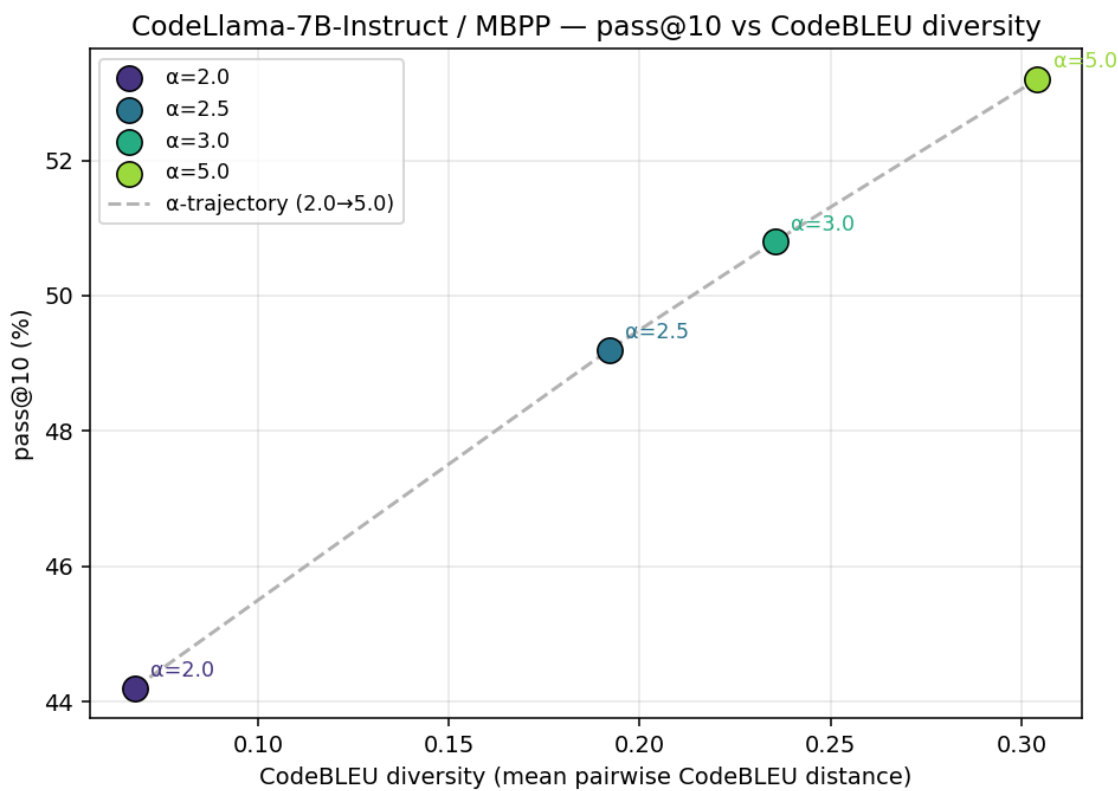


Figure 4: pass@10 vs CodeBLEU diversity

OpenCodeInterpreter-DS-1.3B — HumanEval

n_tasks: 164 · *samples/task*: 10

α	pass@1	pass@3	pass@5	pass@10	codebleu_div	NAUADC
2.0	58.60%	67.98%	71.59%	75.61%	0.1360	1.0362
2.5	56.95%	70.39%	75.01%	79.88%	0.1988	1.0838
3.0	55.85%	69.87%	74.37%	78.66%	0.1998	1.0846
5.0	55.61%	72.20%	77.55%	83.54%	0.2665	1.0947

Source dir: results/pless_alpha_full_humaneval/m-a-p--OpenCodeInterpreter-DS-1.3B/humaneval/metrics/

Files: pless_alpha_a{2.0,2.5,3.0,5.0}_t1.0_metrics.json

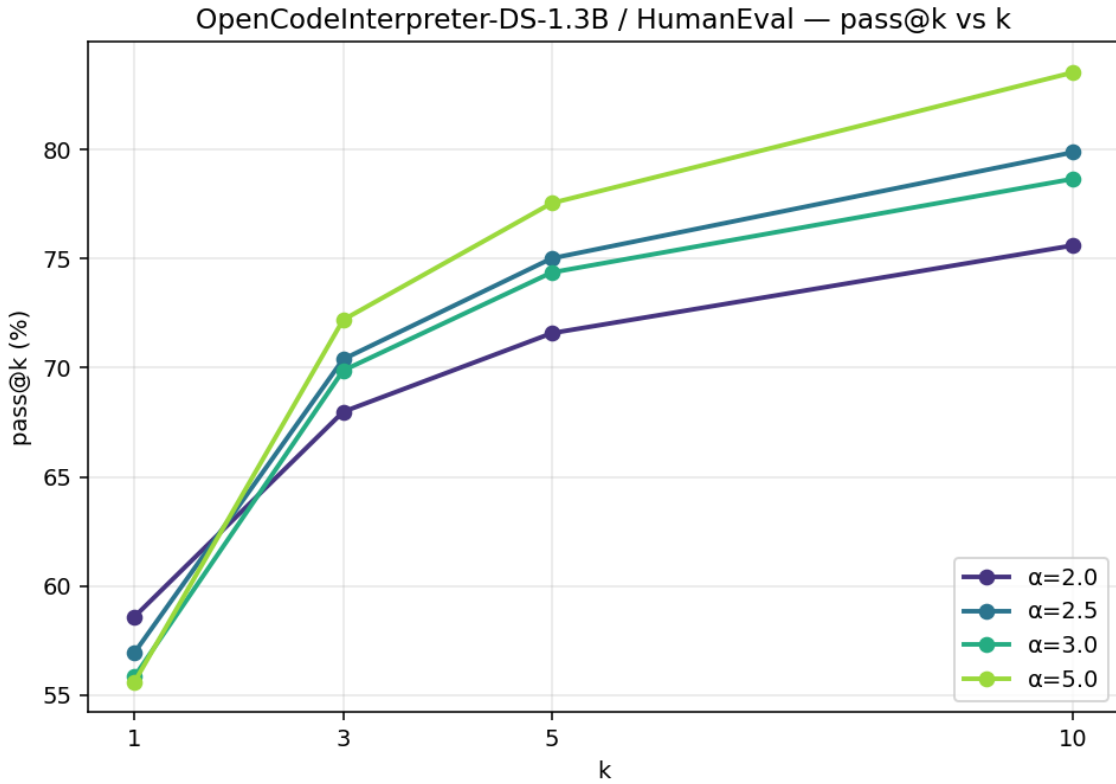


Figure 5: pass@k vs k

OpenCodeInterpreter-DS-1.3B — MBPP

n_tasks: 500 · *samples/task*: 10

α	pass@1	pass@3	pass@5	pass@10	codebleu_div	NAUADC
2.0	47.68%	52.61%	54.25%	55.40%	0.1749	1.0730
2.5	47.28%	55.43%	58.43%	61.40%	0.3515	1.1317
3.0	48.00%	57.95%	61.42%	65.00%	0.4371	1.1655
5.0	46.26%	57.28%	61.40%	66.40%	0.5414	1.2091

Source dir: results/pless_alpha_full_mbpp/m-a-p--OpenCodeInterpreter-DS-1.3B/metrics/

Files: pless_alpha_a{2.0,2.5,3.0,5.0}_t1.0_metrics.json

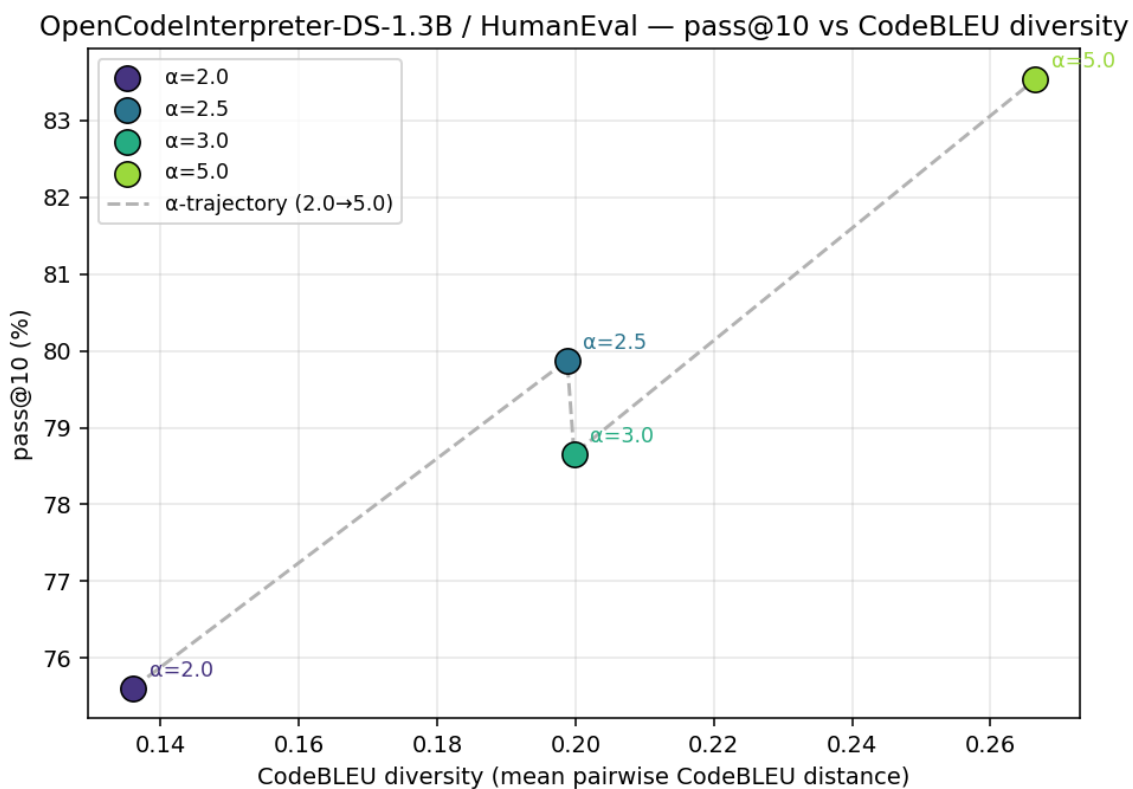


Figure 6: pass@10 vs CodeBLEU diversity

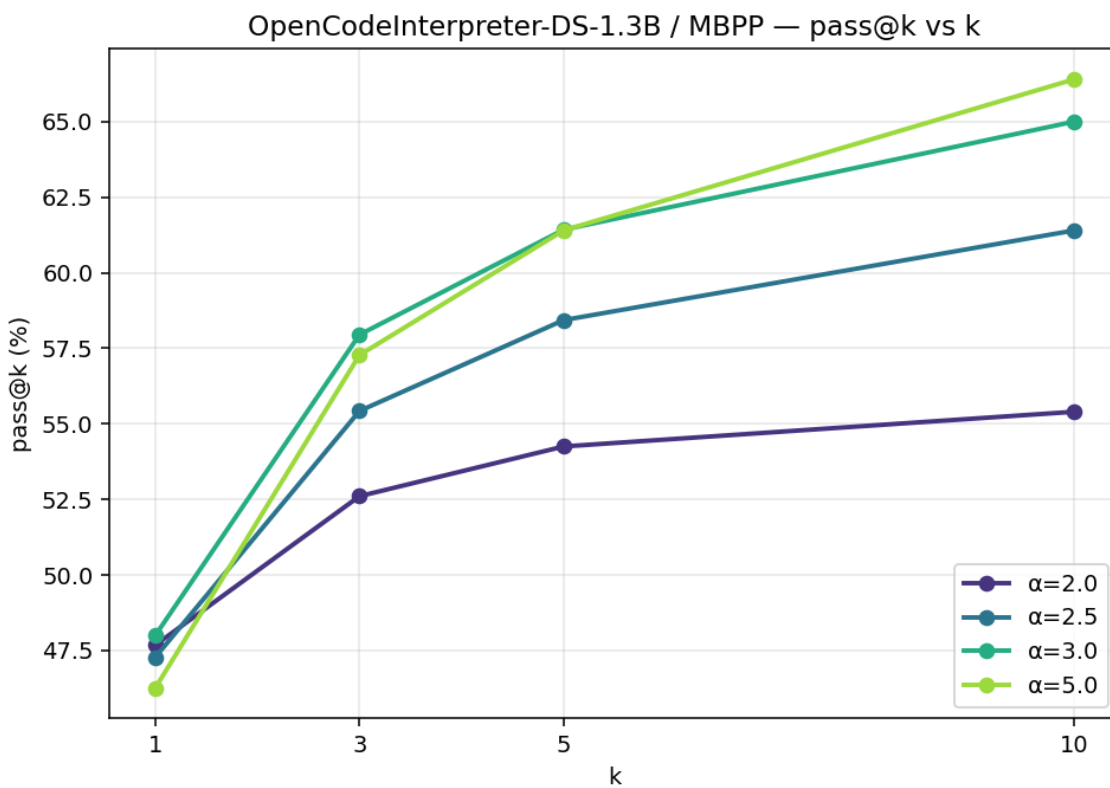


Figure 7: pass@k vs k

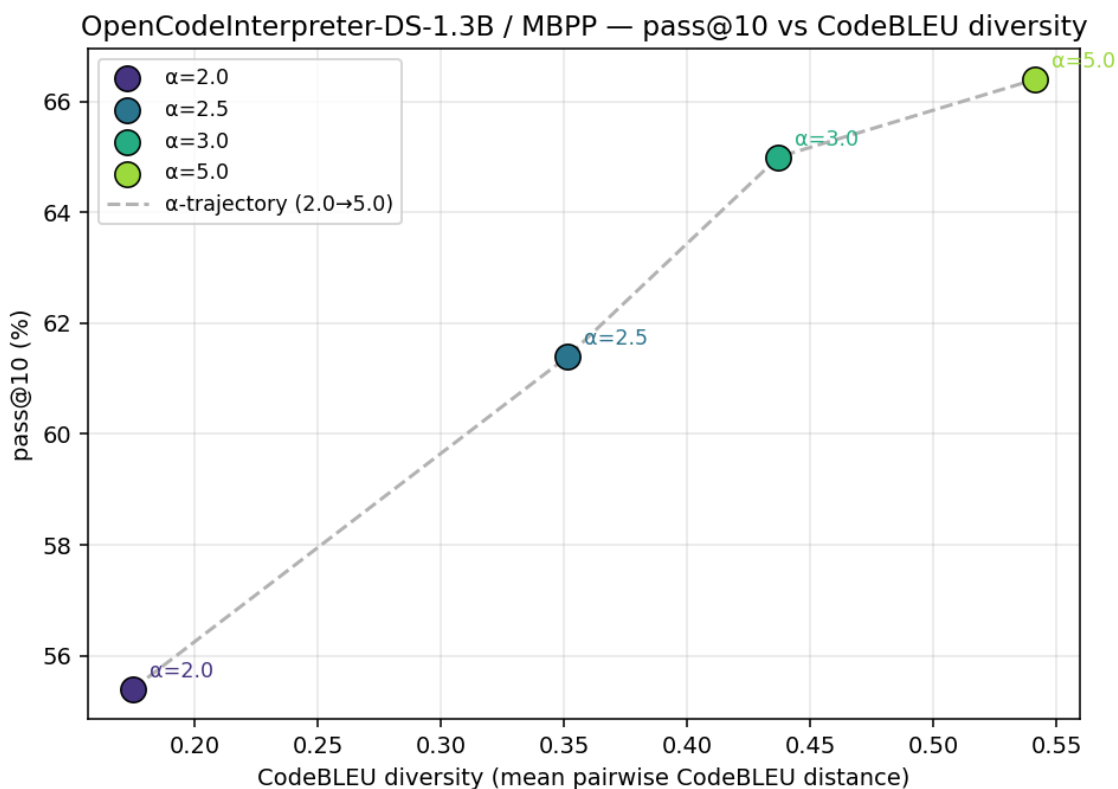


Figure 8: pass@10 vs CodeBLEU diversity

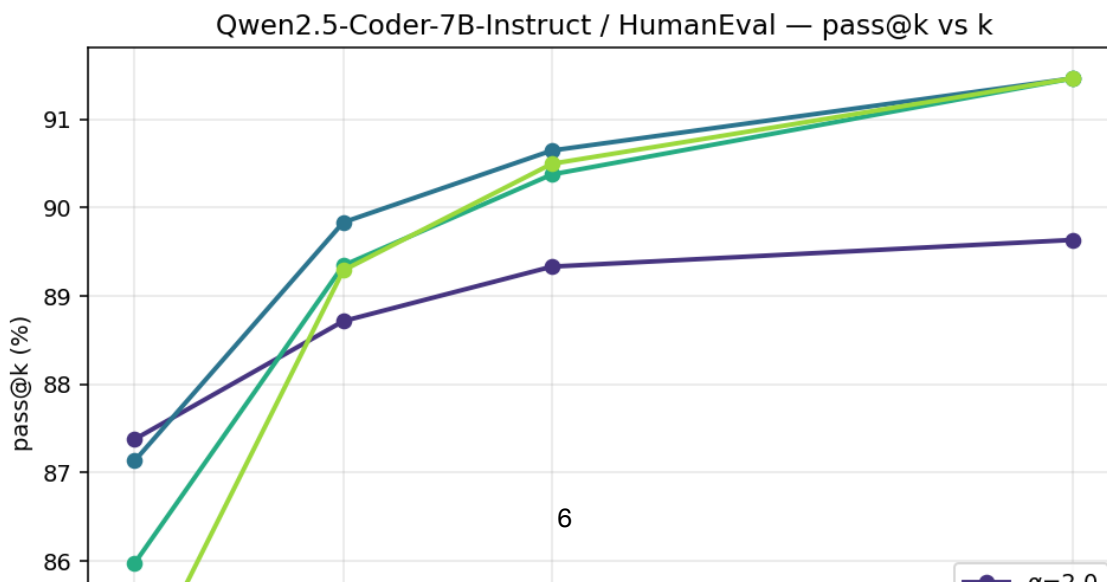
Qwen2.5-Coder-7B-Instruct — HumanEval

n_tasks : 164 · $samples/task$: 10

α	pass@1	pass@3	pass@5	pass@10	codebleu_div	NAUADC
2.0	87.38%	88.71%	89.33%	89.63%	0.0396	1.0000
2.5	87.13%	89.83%	90.65%	91.46%	0.1423	1.0242
3.0	85.98%	89.34%	90.37%	91.46%	0.1693	1.0361
5.0	84.57%	89.29%	90.50%	91.46%	0.2578	1.0648

Source dir: results/pless_alpha_full_humaneval/Qwen--Qwen2.5-Coder-7B-Instruct/humaneval/metrics/

Files: pless_alpha_a{2.0,2.5,3.0,5.0}_t1.0_metrics.json



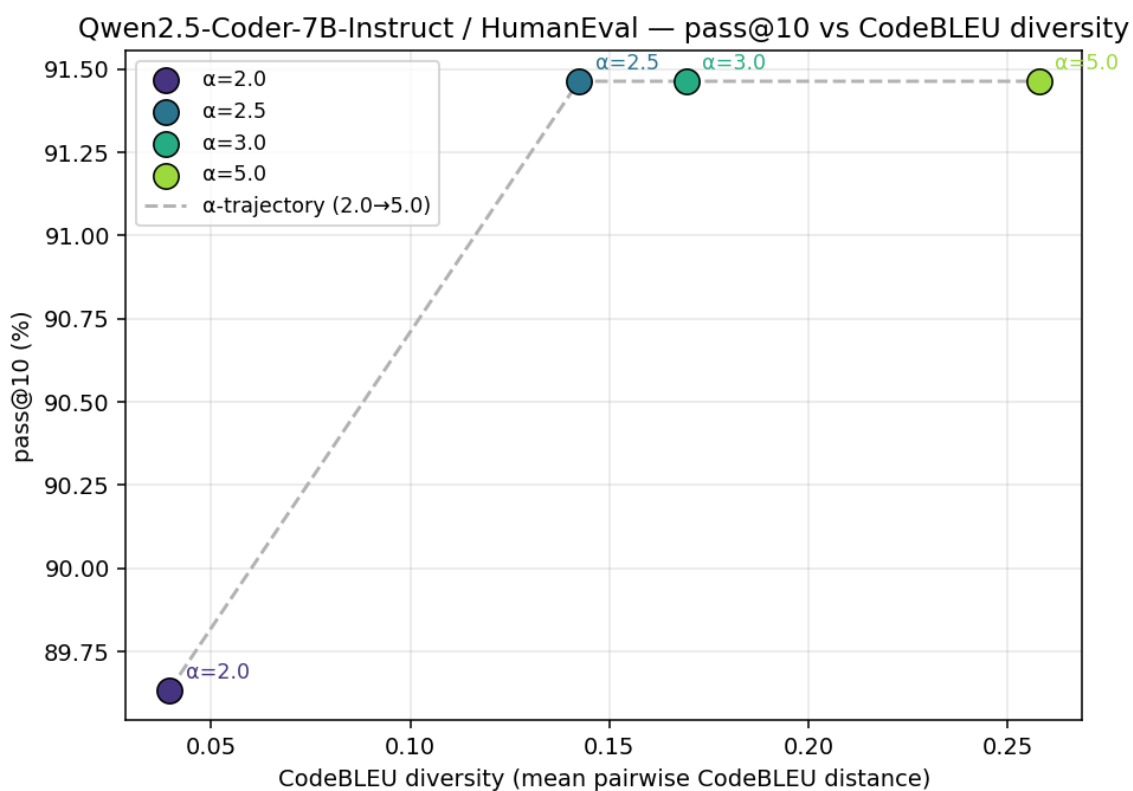


Figure 10: pass@10 vs CodeBLEU diversity

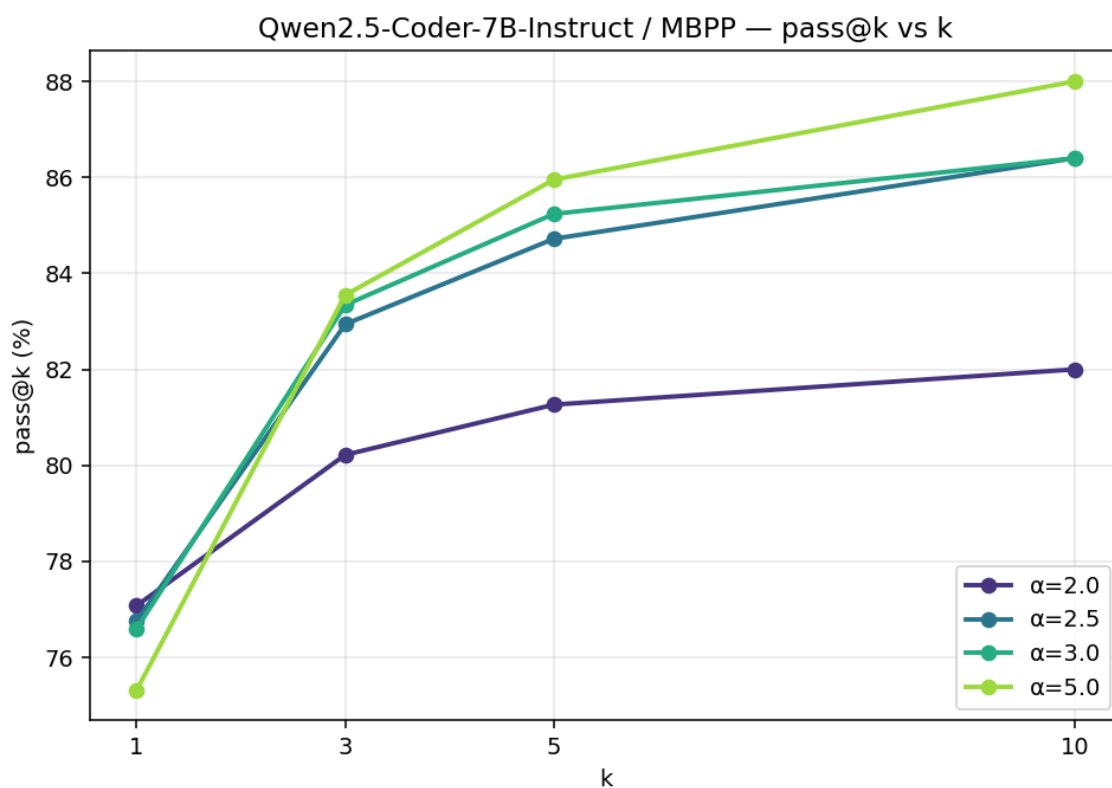


Figure 11: pass@k vs k

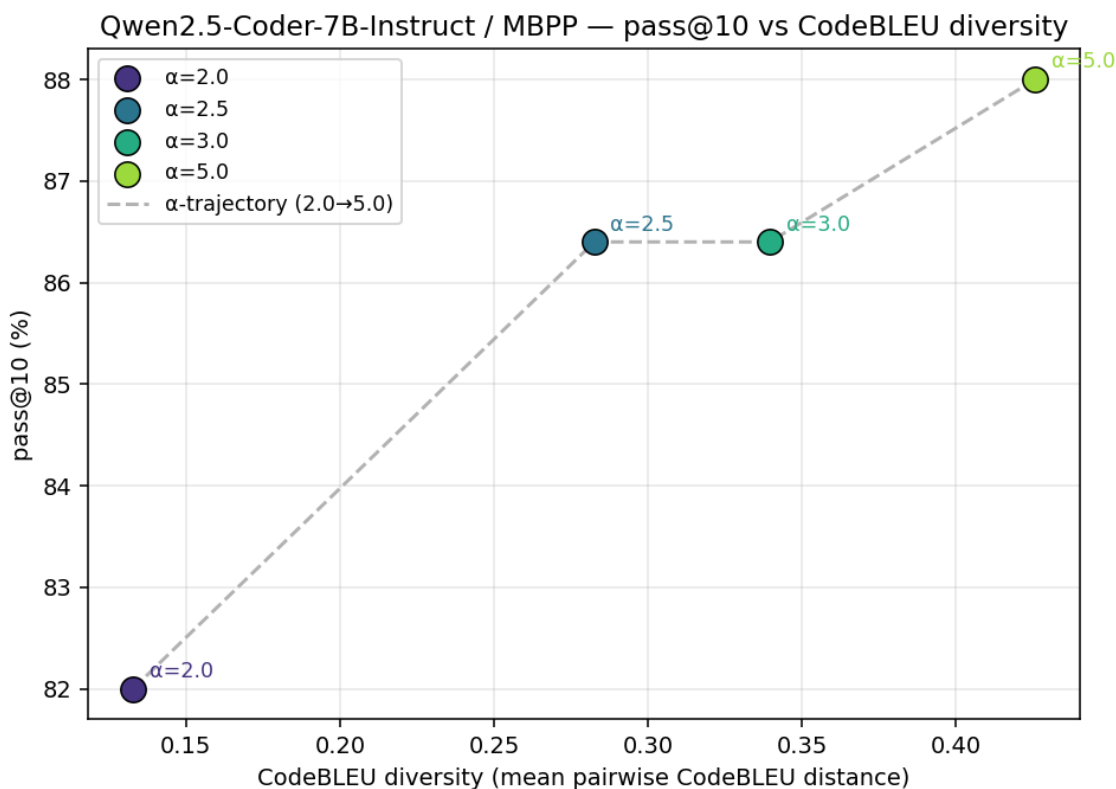


Figure 12: pass@10 vs CodeBLEU diversity

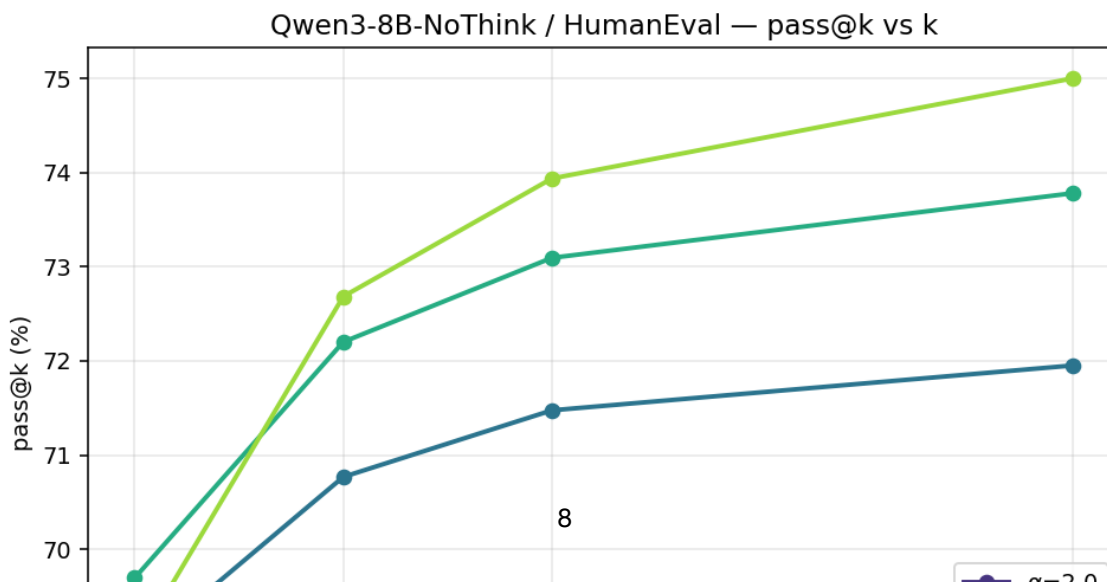
Qwen3-8B-NoThink — HumanEval

n_tasks: 164 · *samples/task*: 10

α	pass@1	pass@3	pass@5	pass@10	codebleu_div	NAUADC
2.0	68.54%	68.94%	69.24%	69.51%	0.0191	1.0000
2.5	68.90%	70.77%	71.47%	71.95%	0.0414	1.0000
3.0	69.70%	72.20%	73.09%	73.78%	0.0671	1.0000
5.0	68.96%	72.68%	73.94%	75.00%	0.0989	1.0073

Source dir: results/pless_alpha_full_humaneval/Qwen--Qwen3-8B/no-think/metrics/

Files: pless_alpha_a{2.0,2.5,3.0,5.0}_t1.0_metrics.json



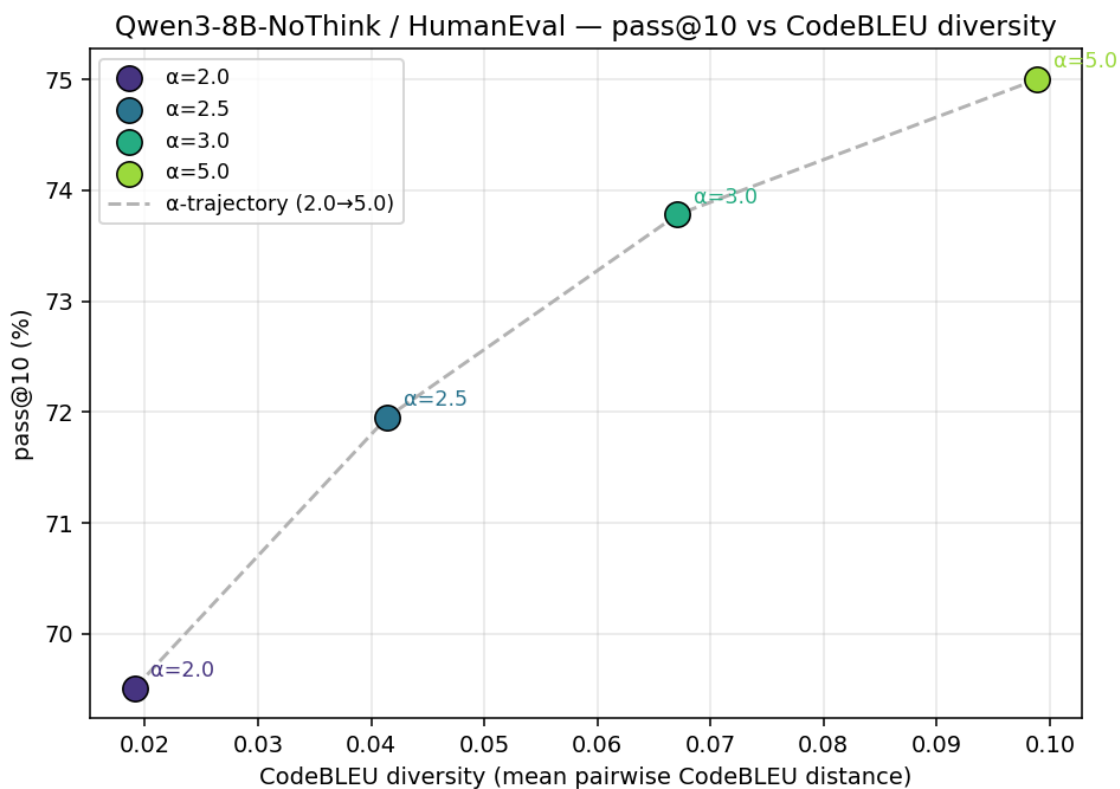


Figure 14: pass@10 vs CodeBLEU diversity

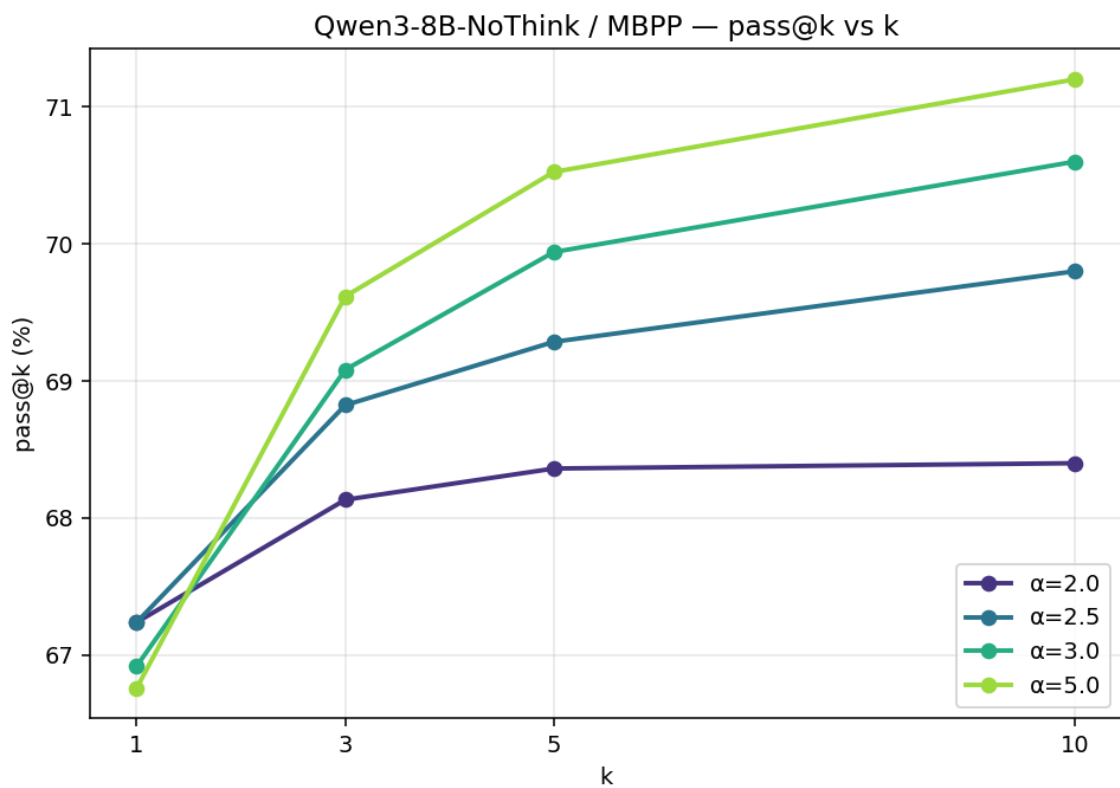


Figure 15: pass@k vs k

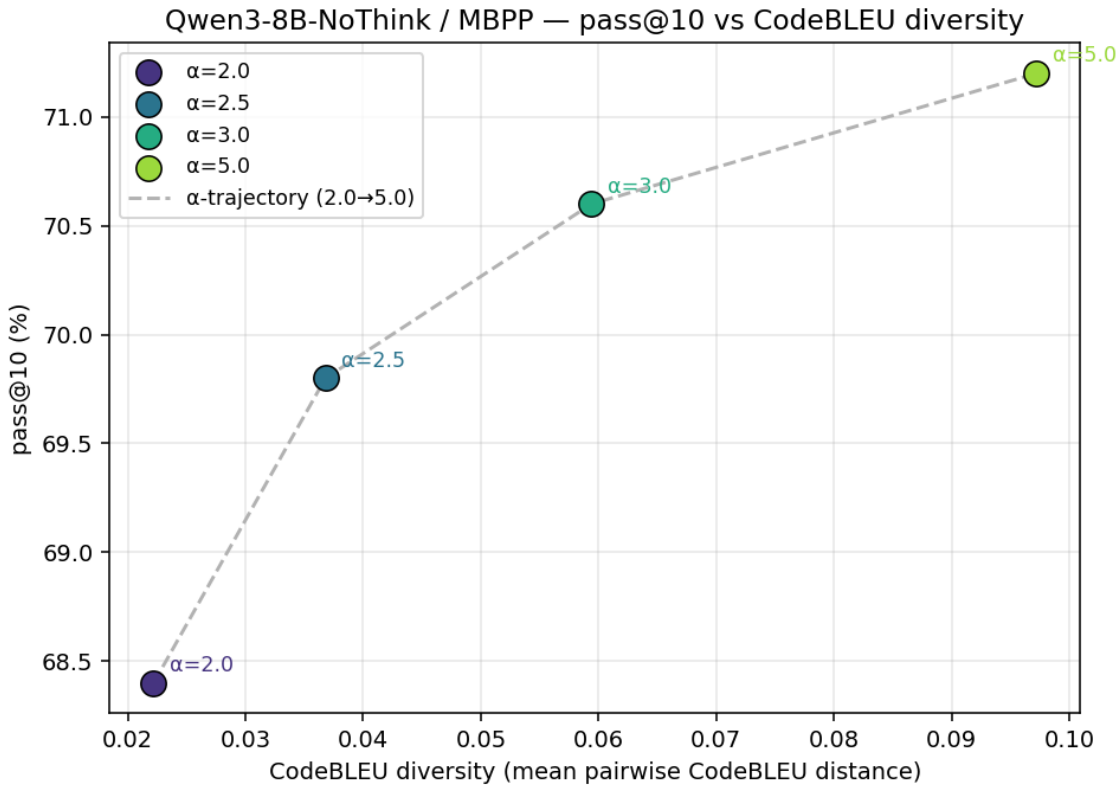


Figure 16: pass@10 vs CodeBLEU diversity

Key qualitative observations

1. **pass@k grows monotonically with k** at every α arm for every (model, dataset) cell — expected from the Chen et al. estimator’s construction; included for completeness.
2. **CodeBLEU diversity grows monotonically with α** at every (model, dataset) cell — see the dashed α -trajectory lines in the scatter plots curving rightward as α grows.
3. **pass@1 mildly decreases with α on the 3 non-thinking models** (typical -1.4 to -3.0 pp from $\alpha=2$ to $\alpha=5$). pass@10 typically grows.
4. **The 3 models occupy distinct operating regimes**: Qwen2.5-Coder is the strongest ($>87\%$ HE pass@10), m-a-p OCI is mid-strength ($>75\%$ HE pass@10 at small size), CodeLlama is the weakest ($<50\%$ on HE).